American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A\,(\overline{B} + C) + \overline{B}(\overline{A} + C) + C(A + B)$$

A. 1
B. 2
C. 3
D. 4
E. 5

2. Boolean Algebra

Simplify the following Boolean algebra expression:

$$\overline{ABC} + \overline{\overline{B} * \overline{A + C}} + \overline{C}(A + B)$$

A. 0
B. $A\overline{C}$
C. $B\overline{C}$
D. $A\overline{C} + B\overline{C}$
E. 1

3. Data Structures

Build a binary search tree for the string (Left subtree ≤ root <right subtree)

CONNECTICUT

A breadth-first traversal is listing each node starting with the root, then each row of the tree from top to bottom. Within each row, nodes are listed from left to right. What is the breadth-first traversal of this tree?

A. CCCNNOTTUEI
B. COCNCTNTEIU
C. CCOCNTNTUEI
D. CCCEINNOTTU
E. COCTNCUTNEI

4. Data Structures

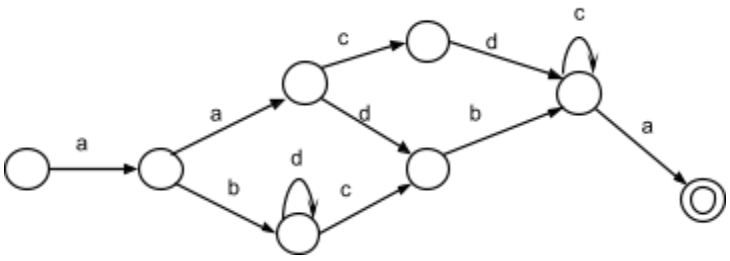
Which of the following strings, when used in the following algorithm on an initially empty stack, will result in a stack that contains the original string?

```
PUSH(first letter in string)
for each other letter in the string
        x = POP()
        PUSH(x)
        if letter != x then
                PUSH(letter)
        end if
next letter
```

A. APPLES
B. CARROT
C. PEPPER
D. ORANGE
E. CHERRY

5. FSAs

Which letter is required to be in all of the accepted strings?



A. a
B. b
C. c
D. d
E. More than 1 of them

6. Regular Expressions

In ACSLland license plates are issued using the following regular expression:

```
[D-R1-5]* [^0-2][a-t]?
```

How many of the following license plates are accepted?

J44ee	EN811f	IR54p	FO795m
PH94n	GM223g	Dan	48ACSL

A. all of them
B. 7
C. half of them
D. 1
E. none of them

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1. 布尔代数

请问有多少个有序三元组可以使得下列布尔表达式的结果为假？

$$A\,(\overline{B}\,+\,C)\,+\overline{B}(\overline{A}\,+\,C)\,+\,C(A\,+\,B)$$

A. 1
B. 2
C. 3
D. 4
E. 5

2. 布尔代数

请简化下列布尔表达式：

$$\overline{ABC}\,+\,\overline{\overline{B}\,*\,\overline{A\,+\,C}}\,+\,\overline{C}(A\,+\,B)$$

A. 0
B. $A\overline{C}$
C. $B\overline{C}$
D. $A\overline{C}\,+\,B\overline{C}$
E. 1

3. 数据结构

在以下字符串构成的二叉搜索树中：（左子树≤根、右子树 > 根）

CONNECTICUT

广度优先遍历是指：从根节点开始遍历，再从上到下依次遍历树的每一层；同一层内的结点按从左到右的顺序遍历。这棵树的广度优先遍历结果是什么？

A. CCCNNOTTUEI
B. COCNCTNTEIU
C. CCOCNTNTUEI
D. CCCEINNOTTU
E. COCTNCUTNEI

4. 数据结构

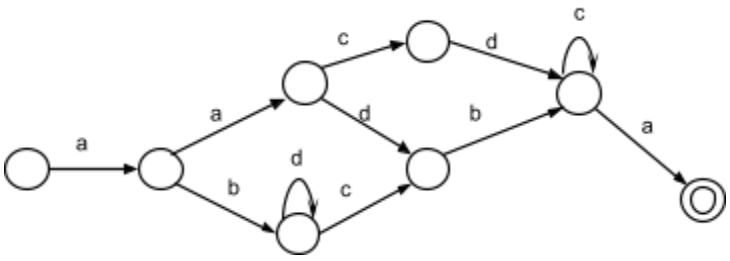
下列字符串中，请问哪一个在初始为空的栈上执行以下算法后，会使栈中包含原始字符串？

```
PUSH(字符串中第一个字母)
for each other letter in the string
    x = POP()
    PUSH(x)
    if letter != x then
        PUSH(letter)
    end if
next letter
```

A. APPLES
B. CARROT
C. PEPPER
D. ORANGE
E. CHERRY

5. FSAs

请问**在所有能被接收的字符串中，必须包含哪个字母？**



A. a
B. b
C. c
D. d
E. 不止**一个字母**

6. 正则表达式

在 ACSL 国，车牌按照下面的正则表达式发放：

```
[D-R1-5]* [^0-2][a-t]?
```

下面的车牌中，有多少个能被该正则表达式匹配？

J44ee	EN811f	IR54p	FO795m
PH94n	GM223g	Dan	48ACSL

A. 所有
B. 7
C. 一半
D. 1
E. 无